

- (vii) Met $L = 1,2 \text{ m}$, $k = 40 \text{ N/m}$ en $m = 0,5 \text{ kg}$, bereken die energie wat in die vere gestoor is wanneer hulle $0,5 \text{ m}$ uitgerek is.
 With $L = 1,2 \text{ m}$, $k = 40 \text{ N/m}$ and $m = 0,5 \text{ kg}$ determine the energy stored in the springs when they are stretched by $0,5 \text{ m}$.

(3)

- (viii) Bepaal die spoed van die massa wanneer dit deur die ewewigsposisie gaan.
 Determine the speed of the mass while it is moving through the equilibrium position.

(3)
[22]

VRAAG / QUESTION 4

- (a) Definieer hoekversnelling α en gee sy eenhede.
 Define angular acceleration and give its units.

$$\alpha = \frac{d\omega}{dt} \quad \text{rad/s}^2$$

(2)

- (b) 'n Wiel met straal r het 'n hoekversnelling α . Skryf 'n uitdrukking neer vir die tangensiale versnelling a_t in terme van α .
 A wheel with radius r has an angular acceleration α . Give an expression for the tangential acceleration a_t in terms of α .

$$a_t = \alpha r \quad \left(a_t = \frac{dv}{dt} = \frac{d(\omega r)}{dt} = r \left(\frac{d\omega}{dt} \right) = r \alpha \right)$$

(1)

- (c) 'n Krag F werk in op 'n liggaam soos in die volgende figuur. Gee 'n uitdrukking vir die draaimoment τ in terme van θ .
 A force F is acting on an object as shown in the following figure. Give an expression for the torque τ in terms of θ .



$$\tau = r F \sin \theta$$

(1)